

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1285

Clifton Forge 138 kV
May 27, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1285. The Interconnection Customer proposes a 109.2 MW battery energy storage facility interconnecting at the Clifton Forge 138 kV substation in the Virginia area. The request was received on September 10, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 373 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,564,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Clifton Forge – Talley Road 138.0 kV
Contingency: Lowmoor – Aspen 345.0 kV
Loading: 127.6% (Rating: 283 MVA)
DFAX: 63.8%
Pre-project loading: 103.0%
2. Facility: Talley Rd Substation – Lowmoor 138.0 kV
Contingency: Aspen – Lowmoor 345.0 kV
Loading: 126.7% (Rating: 290 MVA)
DFAX: 63.8%
Pre-project loading: 102.8%
3. Facility: Craigsville – Yuaipa 138.0 kV
Contingency: Aspen – Lowmoor 345.0 kV
Loading: 121.2% (Rating: 240 MVA)
DFAX: 36.2%
Pre-project loading: 104.7%
4. Facility: Craigsville – Kattelville Substation 138.0 kV Line
Contingency: Aspen – Lowmoor 345.0 kV L
Loading: 119.8% (Rating: 253 MVA)
DFAX: 36.2%
Pre-project loading: 104.2%
5. Facility: Richmnd Q – Kattelville 138.0 kV
Contingency: Aspen – Lowmoor 345.0 kV L
Loading: 119.2% (Rating: 259 MVA)
DFAX: 36.2%
Pre-project loading: 103.9%

6. Facility: Richmond Q – Clifton Forge 138.0 kV L
Contingency: Aspen – Lowmoor 345.0 kV
Loading: 118.0% (Rating: 271 MVA)
DFAX: 36.2%
Pre-project loading: 103.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Talley Road – Clifton Forge 138.0 kV	Rebuild line with higher-capacity conductor	\$9,043,000	69% (\$6,256,000)
Talley Rd Substation – Lowmoor 138.0 kV	Rebuild line with higher-capacity conductor	\$12,350,000	69% (\$8,543,000)
Craigsville – Yuaipa 138.0 kV	Rebuild line with higher-capacity conductor	\$11,251,000	56% (\$6,306,000)
Craigsville – Kattelville Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,895,000	56% (\$3,864,000)
Richmnd Q – Kattelville 138.0 kV	Rebuild line with higher-capacity conductor	\$36,697,000	56% (\$20,568,000)
Clifton Forge – Richmond Q 138.0 kV	Rebuild line with higher-capacity conductor	\$3,617,000	56% (\$2,027,000)
	Total allocated to project		\$47,564,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.